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| <b>DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE</b><br><b>Summer Supplementary Examination – 2022 (Pattern 2017)</b><br><b>Branch : Electronics and Tele Communication Eng.</b><br><b>Course: B. Tech. Semester : IV</b><br><b>Subject Code &amp; Name: BTEXC403 Microprocessor</b><br><b>Max Marks: 60 Date: Duration: 3.00 Hr.</b> |                                                                                                                      |  |          |
| <b>Instructions to the Students:</b><br>1. All questions are compulsory.<br>2. Use of non-programmable scientific calculators is allowed.<br>3. Assume suitable data wherever necessary and mention it clearly.                                                                                                                                 |                                                                                                                      |  |          |
|                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                      |  | Marks    |
| <b>Q. 1</b>                                                                                                                                                                                                                                                                                                                                     | <b>Solve the following.</b>                                                                                          |  |          |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                       | <b>Draw architecture of 8085 microprocessor and explain any three blocks of it.</b>                                  |  | <b>6</b> |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                       | <b>Draw pin diagram of 8085 microprocessor and explain any four pins of it.</b>                                      |  | <b>6</b> |
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| <b>Q.2</b>                                                                                                                                                                                                                                                                                                                                      | <b>Solve any two of the following.</b>                                                                               |  |          |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                       | <b>Explain any three addressing modes of 8085 microprocessor with suitable examples of each.</b>                     |  | <b>6</b> |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                       | <b>Explain following instructions with suitable examples of each.</b><br><i>LXI H, 2005H</i><br><i>MOV A, M</i>      |  | <b>6</b> |
| <b>C)</b>                                                                                                                                                                                                                                                                                                                                       | <b>Explain flag register of 8085 microprocessor.</b>                                                                 |  | <b>6</b> |
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| <b>Q. 3</b>                                                                                                                                                                                                                                                                                                                                     | <b>Solve the following.</b>                                                                                          |  |          |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                       | <b>Explain following instructions with suitable examples of each</b><br><i>RST 0</i><br><i>RST 1</i><br><i>RST 2</i> |  | <b>6</b> |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                       | <b>Define following terms</b><br><i>Vectored Interrupts</i><br><i>Latency Time</i><br><i>Response Time</i>           |  | <b>6</b> |
|                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                      |  |          |
| <b>Q.4</b>                                                                                                                                                                                                                                                                                                                                      | <b>Solve the following.</b>                                                                                          |  |          |
| <b>A)</b>                                                                                                                                                                                                                                                                                                                                       | <b>With neat interfacing diagram, describe interfacing of 4K RAM with 8085 microprocessor.</b>                       |  | <b>6</b> |
| <b>B)</b>                                                                                                                                                                                                                                                                                                                                       | <b>With neat diagram, describe interfacing of 8255 with 8085 microprocessor.</b>                                     |  | <b>6</b> |
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| <b>Q. 5</b> | <b>Solve the following.</b>                                                                                                             |  |          |
| <b>A)</b>   | <b>Explain any three addressing modes of 8086 microprocessor with suitable examples of each.</b>                                        |  | <b>6</b> |
| <b>B)</b>   | <b>Explain following instructions of 8086 microprocessor with suitable examples of each.</b><br><i>SAHF</i><br><i>MUL</i><br><i>SHR</i> |  | <b>6</b> |
|             |                                                                                                                                         |  |          |
|             | <b>*** End ***</b>                                                                                                                      |  |          |